定积分定义与一般性质

$$\lim_{n\to\infty}(\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+\ldots+\frac{1}{\sqrt{n^2+n}})$$
$$\frac{n}{\sqrt{n^2+n}}\le b_n\le\frac{n}{\sqrt{n^2+1}}$$
$$\text{原式}=1$$

$$\lim_{n\to\infty}(\frac{1}{\sqrt{n^2+1^2}}+\frac{1}{\sqrt{n^2+2^2}}+\ldots+\frac{1}{\sqrt{n^2+n^2}})$$
$$\text{原式}=\lim_{n\to\infty}\sum_{i=1}^{n}\frac{1}{\sqrt{n^2+i^2}}=\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n}\frac{1}{\sqrt{(\frac{i}{n})^2+1}}=\int_0^1\frac{dx}{\sqrt{x^2+1}}=\ln(x+\sqrt{x^2+1})|_0^1=\ln(1+\sqrt{2})$$

$$\lim_{n\to\infty}(\frac{\sqrt{n^4-i^4}}{n^4}+\frac{2\sqrt{n^4-2^4}}{n^4}+\ldots+\frac{n\sqrt{(n^4-n^4)}}{n^4})$$
$$\text{原式}=\lim_{n\to\infty}\sum_{i=1}^{n}\frac{i\sqrt{n^4-i^4}}{n^4}=\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n}\frac{i}{n}\sqrt{1-(\frac{i}{n})^4}=\int_0^1 x\sqrt{1-x^4}dx$$

$$\lim_{n\to\infty}(\frac{1}{n^2+1^2}+\frac{2}{n^2+2^2}+\ldots+\frac{n}{n^2+n^2})$$
$$\text{原式}=\lim_{n\to\infty}\sum_{i=1}^{n}\frac{i}{n^2+i^2}=\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n}\frac{\frac{i}{n}}{1+(\frac{i}{n})^2}=\int_0^1\frac{x}{1+x^2}dx=\frac{1}{2}\ln 2$$

$$-|f(x)|\le f(x)\le|f(x)|$$
$$-\int_a^b|f(x)|dx\le\int_a^b f(x)dx\le\int_a^b|f(x)|dx$$
$$-B\le A\le B$$
$$|\int_a^b f(x)dx|\le\int_a^b|f(x)|dx$$

$$\text{积分中值定理}$$
$$f(x)\in C[a,b],\exists\xi\in[a,b]$$
$$\int_a^b f(x)dx=f(\xi)(b-a)$$

$$f(x)\in[a,b]\Rightarrow\exists m,M,\text{使}m\le f(x)\le M$$
$$\int_a^b mdx\le\int_a^b f(x)dx\le\int_a^b Mdx$$
$$m(b-a)\le\frac{1}{b-a}\int_a^b f(x)dx\le M(b-a)$$
$$\exists\xi\in[a,b],\text{使}f(\xi)=\frac{1}{b-a}\int_a^b f(x)dx$$

$$f(x)\in C[0,1],(0,1)\text{可导},f(1)=4\int_0^{\frac{1}{4}}f(x)dx$$
$$\text{证}:\exists\xi\in(0,1),\text{使}f'(\xi)=0$$
$$f(x)\in C[0,\frac{1}{4}]\Rightarrow\exists c\in[0,\frac{1}{4}],\text{使}$$
$$\int_0^{\frac{1}{4}}f(x)dx=f(c)(\frac{1}{4}-0)\Rightarrow 4\int_0^{\frac{1}{4}}f(x)dx=f(c)$$
$$\Rightarrow f(c)=f(1)$$
$$\exists\xi\in(c,1)\subset(0,1),\text{使}f'(\xi)=0$$

积分基本定理与特殊性质

$$\text{积分上限函数}$$
$$\Phi(x)=\int_a^x f(x)dx=\int_a^x f(t)dt$$
$$\int_a^x f(x,t)dt\text{在表达式中的}x\text{与上限}x\text{同}$$

$$设f(x)\in C[a,b],\Phi(x)=\int_a^x f(t)dt则$$

$$\Phi'(x)=\frac{d}{dx}\int_a^x f(t)dt=f(x)$$

$$证:\Delta\Phi=\Phi(x+\Delta x)-\Phi(x)=\int_a^{x+\Delta x}f(t)dt-\int_a^x f(t)dt=\int_x^{x+\Delta x}f(t)dt$$

$$\because f(x)\in C[a,b],\therefore \Delta\Phi=f(\xi)\Delta x\ (\xi在x与x+\Delta x之间)$$

$$\Rightarrow\frac{\Delta\Phi}{\Delta x}=f(\xi)\Rightarrow\lim_{\Delta x\to 0}\frac{\Delta\Phi}{\Delta x}=\lim_{\Delta x\to 0}f(\xi)$$

$$\because f(x)连续,\therefore\lim_{\Delta x\to 0}f(\xi)=\lim_{\xi\to x}f(\xi)=f(x)$$

$$\Rightarrow\Phi'(x)=f(x)$$

$$\frac{d}{dx}\int_a^{\phi(x)}f(t)dt=f[\phi(x)]\phi'(x)$$

$$f(x)连续,f(0)=0,f'(0)=\pi,求\lim_{x\to 0}\frac{\int_0^x f(t)dt}{x-\ln(1+x)}$$

$$原式=\lim_{x\to 0}\frac{f(x)}{1-\frac{1}{x+1}}=\lim_{x\to 0}\frac{(x+1)f(x)}{x}=\lim_{x\to 0}\frac{f(x)+(x+1)f'(x)}{1}=f(0)+f'(0)=\pi$$

$$设函数f(x)连续,\phi(x)=\int_0^x(x-t)f(t)dt,求\phi''(x)$$

$$\phi(x)=x\int_0^x f(t)dt-\int_0^x tf(t)dt$$

$$\phi'(x)=\int_0^x f(t)dt+xf(x)-xf(x)$$

$$\phi''(x)=f(x)$$

$$\int_a^b f(x)dx=\lim_{\lambda\to 0}\sum_{i=1}^n f(\xi_i)\Delta x_i$$

牛顿莱布尼茨公式

$$f(x)\in C[a,b],F'(x)=f(x)$$

$$则\int_a^b f(x)dx=F(b)-F(a)$$

$$证:\Phi(x)=\int_a^x f(t)dt,\Phi'(x)=f(x)$$

$$\because F'(x)=f(x),\therefore F(x)-\Phi(x)=C_0(a\le x\le b)$$

$$\Rightarrow F(a)-\Phi(a)=F(b)-\Phi(b)$$

$$\because\Phi(a)=0,\therefore\Phi(b)=F(b)-F(a)$$

$$即\int_a^b f(t)dt=F(b)-F(a)或\int_a^b f(x)dx=F(b)-F(a)$$

$$\int_0^1\frac{x}{1+x^4}dx=\frac{1}{2}\int_0^1\frac{d(x^2)}{1+(x^2)^2}=\frac{1}{2}\arctan x^2\ |_0^1=\frac{\pi}{8}$$

换元积分法

$$\begin{aligned}
&\int_0^4\frac{x+1}{\sqrt{2x+1}}dx\\
=&\frac{1}{4}\int_0^4\frac{(2x+1)+1}{\sqrt{2x+1}}d(2x+1)\\
=&\frac{1}{4}\int_0^4(2x+1)^{\frac{1}{2}}d(2x+1)+\frac{1}{2}\int_0^4\frac{d(2x+1)}{2\sqrt{2x+1}}\\
=&\frac{1}{4}*\frac{2}{3}(2x+1)^{\frac{3}{2}}\ |_0^4+\frac{1}{2}\sqrt{2x+1}\ |_0^4\\
=&\frac{16}{3}
\end{aligned}$$

$$\int_0^4 \frac{x+1}{\sqrt{2x+1}}dx$$
$$\text{令}\sqrt{2x+1}=t \Rightarrow x=\frac{1}{2}(t^2-1)$$
$$=\int_1^3 \frac{\frac{1}{2}(t^2+1)}{t} * tdt$$
$$=\int_1^3 \frac{1}{2}(t^2+1)dt$$
$$=\frac{1}{2} * (\frac{t^3}{3} \mid_1^3 +2)$$
$$=\frac{16}{3}$$

$$\int_0^2 xe^{-x^2}dx=\frac{1}{2}\int_0^2 e^{-x^2}d(x^2)=\frac{1}{2}\int_0^4 e^{-x}d(x)=-\frac{1}{2}e^{-x} \mid_0^4=\frac{1}{2}(1-e^{-4})$$

$$\text{设}f(x)\text{连续}, f(0)=0, f'(0)=4, \text{求}\lim_{x\to 0}\frac{\int_0^x tf(x^2-t^2)dt}{x^4}$$

$$\int_0^x tf(x^2-t^2)dt=-\frac{1}{2}\int_0^x f(x^2-t^2)d(-t^2)=-\frac{1}{2}\int_0^x f(x^2-t^2)d(x^2-t^2)=-\frac{1}{2}\int_{x^2}^0 f(u)du=\frac{1}{2}\int_0^{x^2} f(u)du$$

$$\lim_{x\to 0}\frac{\int_0^x tf(x^2-t^2)dt}{x^4}=\frac{1}{4}\lim_{x\to 0}\frac{f(x^2)}{x^2}=\frac{1}{4}\lim_{x\to 0}\frac{f(x^2)-f(0)}{x^2}=\frac{1}{4}f'(0)=1$$

分部积分法

$$(uv)'=u'v+uv'$$
$$\Rightarrow uv \mid_a^b=\int_a^b vdu+\int_a^b udv$$
$$\Rightarrow \int_a^b udv=uv \mid_a^b-\int_a^b vdu$$

$$\int_0^1 x\ln(1+x)dx$$
$$=\int_0^1 \ln(1+x)d(\frac{1}{2}x^2)$$
$$=\frac{x^2}{2}\ln(1+x) \mid_0^1-\frac{1}{2}\int_0^1 \frac{(x^2-1)+1}{x+1}dx$$
$$=\frac{1}{2}\ln 2-\frac{1}{2}\int_0^1 (x-1+\frac{1}{x+1})dx$$
$$=\frac{1}{2}\ln 2-\frac{1}{2}(\frac{1}{2}-1+\ln 2)$$
$$=\frac{1}{4}$$

特殊性质

对称区间

$$\text{设}f(x)\in C[-a,a], \text{则}\int_{-a}^a f(x)dx=\int_0^a [f(x)+f(-x)]dx$$
$$\text{证：左}=\int_{-a}^0 f(x)dx+\int_0^a f(x)dx$$
$$\text{而}\int_{-a}^0 f(x)dx=\int_{-a}^0 f(-t)d(-t)=\int_0^a f(-t)dt=\int_0^a f(-x)dx$$
$$\text{左}=\int_0^a [f(x)+f(-x)]dx$$
$$\text{若}f(-x)=-f(x)\Rightarrow\int_{-a}^a f(x)dx=0$$
$$\text{若}f(-x)=f(x)\Rightarrow\int_{-a}^a f(x)dx=2\int_0^a f(x)dx$$

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{e^x+1}dx=\int_0^{\frac{\pi}{4}}(\frac{1}{e^x+1}+\frac{1}{e^{-x}+1})dx=\int_0^{\frac{\pi}{4}}(\frac{1}{e^x+1}+\frac{e^x}{e^x+1})dx=\frac{\pi}{4}$$

$$I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{1-\sin x}dx=\int_0^{\frac{\pi}{4}}(\frac{1}{1-\sin x}+\frac{1}{1+\sin x})dx=\int_0^{\frac{\pi}{4}} \frac{2}{1-\sin^2 x}=2\int_0^{\frac{\pi}{4}} \sec^2 xdx=2\tan x \mid_0^{\frac{\pi}{4}}=2$$

三角函数

$$\text{设}f(x)\in C[0,1],\text{则}\int_0^{\frac{\pi}{2}}f(\sin x)dx=\int_0^{\frac{\pi}{2}}f(\cos x)dx$$

$$x=-t,\int_0^{-a}=\int_0^{a}$$

$$x+t=a+b,\int_a^{b}=\int_a^{b}$$

$$\text{证}:x+t=\frac{\pi}{2},d(x+t)=dx+dt=0$$

$$\int_0^{\frac{\pi}{2}}f(\sin x)dx=\int_{\frac{\pi}{2}}^{0}f(\cos t)(-dt)=\int_0^{\frac{\pi}{2}}f(\cos t)dt$$

$$\int_0^{\frac{\pi}{2}}f(\sin x)dx=\int_0^{\frac{\pi}{2}}f(\cos x)dx$$

$$I=\int_0^{\frac{\pi}{2}}\frac{\sin x}{\sin x+\cos x}dx$$

$$I=\int_0^{\frac{\pi}{2}}\frac{\cos x}{\cos x+\sin x}dx$$

$$2I=\int_0^{\frac{\pi}{2}}1dx=\frac{\pi}{2}$$

$$I=\frac{\pi}{4}$$

$$I=\int_0^{\frac{\pi}{2}}\sin^2 xdx$$

$$I=\int_0^{\frac{\pi}{2}}\cos^2 xdx$$

$$2I=\int_0^{\frac{\pi}{2}}1dx=\frac{\pi}{2}$$

$$I=\frac{\pi}{4}$$

$$\text{记}:\int_0^{\frac{\pi}{2}}\sin^n xdx=\int_0^{\frac{\pi}{2}}\cos^n xdx=I_n$$

$$\begin{cases}I_n=\frac{n-1}{n}I_{n-2}\\I_0=\frac{\pi}{2}\\I_1=1\end{cases}$$

$$\int_0^{\frac{\pi}{2}}\sin^{10}x=I_{10}=\frac{9}{10}I_8=\frac{9}{10}*\frac{7}{8}*I_6=\frac{9}{10}*\frac{5}{6}*\frac{3}{4}*\frac{1}{2}*I_0=\frac{9}{10}*\frac{5}{6}*\frac{3}{4}*\frac{1}{2}*\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}}\cos^{11}x=I_{11}=\frac{10}{11}*\frac{8}{9}*\frac{6}{7}*\frac{4}{5}*\frac{2}{3}*I_1=\frac{10}{11}*\frac{8}{9}*\frac{6}{7}*\frac{4}{5}*\frac{2}{3}*1$$

$$I=\int_0^{\frac{\pi}{2}}\sin^8 x\cos^2 xdx$$

$$I=\int_0^{\frac{\pi}{2}}\sin^8 x(1-\sin^2 x)dx=\int_0^{\frac{\pi}{2}}\sin^8 xdx-\int_0^{\frac{\pi}{2}}\sin^{10}xdx=\frac{7}{8}*\frac{5}{6}*\frac{3}{4}*\frac{1}{2}*I_0-\frac{9}{10}*\frac{7}{8}*\frac{5}{6}*\frac{3}{4}*\frac{1}{2}I_0=\frac{1}{10}*\frac{7}{8}*\frac{5}{6}*\frac{3}{4}*\frac{1}{2}$$

$$\int_0^1 x^3\sqrt{1-x^2}dx=\int_0^{\frac{\pi}{2}}\sin^3 t*\cos^2 tdt=\int_0^{\frac{\pi}{2}}\sin^3 t*(1-\sin^2 t)dt=I_3-I_5$$

$$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{\sin^4 x}{1+e^{-x}}dx=\int_0^{\frac{\pi}{2}}(\frac{\sin^4 x}{1+e^{-x}}+\frac{\sin^4 x}{1+e^{x}})dx=\int_0^{\frac{\pi}{2}}\sin^4 x(\frac{e^x}{1+e^x}+\frac{1}{1+e^x})dx=\int_0^{\frac{\pi}{2}}\sin^4 xdx=I_4=\frac{3}{4}*\frac{1}{2}*I_0=\frac{3\pi}{16}$$